1. Introduction and Exploratory Data Analysis

1.1 Introduction

The objective of this report is to estimate the price elasticity of demand for passenger rail tickets. By leveraging booking data and controlling for observable temporal, product and customer characteristics, we estimate the conditional relationship between ticket price and recorded ticket demand.

1.2 Data and cleaning

The dataset records rail ticket sales made between June 2018 and June 2019, for departures running to June 2020 on 15 anonymised train services totalling 209,697 rows. A row is a sales record that may represent multiple seats, with mean_net_ticket_price recording the average net fare for the seats represented by that record. Each record contains the following observed characteristics:

- Scheduling: Train number, purchase date, and departure date (from which lead time is derived)
- Ticket Type: Cabin class, trip direction (inbound/ outbound), and travel nature (one-way/ round trip).
- Demographics: Customer category

One important derived variable is the lead time. It is the gap between the purchase date and the departure date, ranging from 0 to 364 days (median 39), which drives the sample rule in 1.3 and the tiers in 1.5.One feature of the grouping does matter: the data is not unique. 3,407 attribute combinations appear in more than one row at different fares (median gap 77), and with no fare-class or time-of-day field, those repeats are the only visible trace of intra-day fare tiers. This unobserved heterogeneity may contribute to the measurement error in P.

1.3 Demand and price distributions

Exploratory analysis reveals that ticket price and transaction volume are highly right-skewed. We applied a natural logarithmic transformation to mitigate the influence of extreme outliers and allow the coefficient on log price to be interpreted as price elasticity of demand.

1.4 Price-Quantity relationship

A simple bivariate plot between price and number of tickets sold indicates a negative relationship with a slope of -0.306 as shown by *Figure 1*. However, the baseline estimate is heavily confounded by other omitted variables.

1.5 Temporal Dynamics and Advance Purchase Windows

Booking lead time is calculated as the number of days between purchase and departure. Based on *Figure 2* where lead times are grouped into 10 day buckets, mean prices fall consistently from $403 to roughly $130 by day 120, after which they fluctuate randomly.Concurrently, ticket volume shifts nonmonotonically. If price alone drives demand, quantity should trend inversely across lead time buckets. This mismatch indicates that temporal factors also influence demand. To reduce noise, we restricted our sample to a 120-day booking window.

Furthermore, structural breaks in ticket pricing and volume emerge at the 30-, 14-, 7-, and 2-day pre-departure marks, as seen in *Figure 3* . To prevent supply-side algorithmic adjustments from biasing our elasticity estimates, we defined fixed-effects lead-time bins around these exact windows (0–2, 3–7, 8–14, 15–30, and >30 days). Lacking universal regulatory standards, we validated these cutoffs against peer-reviewed literature:

- **Consumer Behavior:** Yuan, Ran, and Zhang (2026)[1] utilize these identical intervals to segment travel utility, finding that price-elastic leisure travelers book 15–30 days in advance, whereas inelastic business travelers dominate the 0–7 day windows.
- **Algorithmic Pricing:** Hotle, Castillo, Garrow, and Higgins[2] demonstrate that revenue management systems explicitly tie discount expirations to 3-, 7-, and 14-day deadlines to execute peak-load pricing.

1. 6 Market Heterogeneity and Fixed Effects

- Average prices and quantities differ across train identifiers. Mean prices range from around $200 for train E to around $652 for train O and mean quantity ranges from 1.0 to 2.8 seats/record as shown in *Figure 4*. (Train O contains only one record, which is addressed in the sensitivity analysis). These differences could reflect factors such as route and train timings which cannot be inferred from the dataset.

- Special and normal cabins have >2x price gap, with normal cabins driving higher sales volume as shown in *Figure 5*. Additionally, isReturn and isOneway represent distinct ticket dimensions rather than mutually exclusive alternatives. All four binary combinations were present in the data, requiring independent categorical controls.
- Customers in Category A average higher prices ($356) compared to Customer Category B ($194), though Category B purchases more seats per transaction (*Figure 6*).
- Demand varies systematically across departure months and days of the week (*Figure 7, Figure 8*), reflecting unobserved seasonal and scheduling preferences.

2. Empirical Strategy

2.1 Sample Selection

Purchase dates in the dataset span only 1 June 2018 to 30 June 2019, while departures run to June 2020 and observed lead times extend to 364 days. Departures in the second half of 2018 are therefore observed only over their late, expensive bookings, and departures after June 2019 only over their early, cheap ones. Because the two truncations run in opposite directions, pooling them does not cancel out: it produces a spurious "high price, low quantity" pattern at one end of the sample and a "low price, low quantity" pattern at the other, neither of which reflects demand behaviour.

Our sample rule removes this differential truncation by making the observation window identical across departures: we keep records whose departure date falls between 29 September 2018 and 30 June 2019 and whose lead time is at most 120 days. Every retained departure is then observed over exactly the same interval, [0, 120] days before departure. Restricting lead time alone would not achieve this — a July 2018 departure is only ever observable over its final few weeks — so the departure-date restriction is what makes the comparison valid.

Table 1 in the appendix shows how window L was chosen. We track the number of records the regression can use and the share of the booking curve observed within L days of departure. Sample size peaks between 90 and 120 days: moving from 90 to 120 costs 579 records (0.5%) and buys 8.0 percentage points of coverage, roughly 72 records per point. Beyond 120 the trade-off turns sharply — reaching 92.6% coverage at L = 180 costs 17,365 records, about 1,754 records per point, and L = 240 costs 5,287 per point. At L = 120 we therefore retain a near-maximal sample of 123,991 records while observing over 80% of the booking curve, and we adopt it as the analysis sample.

The restriction works as intended. The ratio between the highest and lowest monthly observed mean fare falls from 4.18 to 1.36, and the standard deviation of monthly mean lead time falls from 112.4 days to 3.9 days, confirming that departures in the analysis sample are observed at comparable points on the booking curve.

Two costs follow. The sample retains 123,991 of 209,697 records (59.1%) and spans nine months, so no year-on-year comparison is possible and the full seasonal cycle is not covered. More importantly, the estimand changes: our elasticity describes demand within 120 days of departure, not over the entire selling period. We considered reweighting the full sample by coverage instead, but rejected it because it assumes one month's lead-time distribution holds in every other month — the very assumption the truncation makes untestable.

2.2 Main Specification

Using the 120-day analysis sample defined above, we estimate a log-log ordinary least squares (OLS) specification at the sales-record level:

$$ln(Q_i) = \alpha + \beta ln(P_i) + \gamma' X_i + \delta_{L(i)} + \varepsilon_i$$

where Q_i represents the total seats transacted (*num_seats_total)*, P_i denotes the average net ticket price (*mean_net_ticket_price)*, and X_i is a vector of fixed-effect controls including train identifiers, cabin type, inbound/ outbound direction, return/ one-way indicators, customer category, departure day-of-week, and departure year-month. $\delta_{L(i)}$ captures the effect associated with the booking lead-time category of observation i, based on the

five temporal windows defined in Section 1. Reference Model 0 utilizes the exact same specification but omits the lead-time indicator($\delta_{L(i)}$). Model 1 introduces these booking-stage controls and serves as our main specification. This comparison allows us to examine whether accounting for booking timing materially changes the estimated price–quantity relationship. Model 1 retains exact observed prices. Grouping observations by exact price and the relevant record characteristics resulted in no additional aggregation. All 123,991 observations are preserved, maximizing the retention of intra-day price variation and using coarser price grouping only as a sensitivity analysis.

2.3 Estimation and Robustness

All models are estimated using OLS with HC3 heteroskedasticity-robust standard errors. Residual diagnostics and the Breusch–Pagan test indicate non-constant residual variance, motivating robust inference. HC3 changes the estimated standard errors and confidence intervals but not the OLS coefficient estimates.

We conduct two main robustness exercises to validate our findings:

- Test 1 estimates Model 1 separately for Customer Categories A and B, with additional checks removing the lead-time controls and the rare high-leverage train observation.
- Test 2 applies alternative $10, $20, $50 and $100 price bins. Because binning combines previously distinct records, this alters both price grouping and the effective aggregation of quantity. Test 2 therefore serves as a sensitivity check to alternative price-grouping and cell-aggregation choices, rather than an alternative preferred specification.

As an additional robustness check, we re-estimated Model 1 with standard errors clustered by train, which allows for correlated errors within the same train service. The price coefficient remains significant ($p < 0.001$) under this specification.

3. Results and Interpretation

3.1 Baseline Demand Model

Model 0 provides a baseline estimate of price elasticity without controlling for the booking curve. Using 123,991 observations, the estimated coefficient on $ln(P)$ is −0.207 (HC3 SE = 0.005, $p < 0.001$). Since both price and quantity are log-transformed, it indicates 1% higher ticket price is associated with approximately 0.207% decrease in seats sold, holding train identifier, ticket type and customer category and departure characteristics constant.

Model 1 extends Model 0 by introducing booking lead time effects. Upon controlling for the booking window, the price elasticity coefficient becomes −0.253 (HC3 SE = 0.005, $p < 0.001$), and Adjusted R^2 increases from 0.128 to 0.136.The coefficient becomes more negative (from -0.207 to -0.253) once lead-time controls are added , which suggests that booking timing is an important factor in the observed relationship between price and demand. Model 1 is therefore retained as the preferred main specification.

Refer to Figure 9 in the Appendix for the relevant results generated using Python.

3.2 Heterogeneity by Customer Category

When Model 1 is estimated on split samples by customer category, the price coefficient diverges significantly: +0.165 ($p < 0.001$) for Customer A and −0.296 ($p < 0.001$) for Customer B. The positive coefficient for Customer A should not be interpreted as a violation of the law of demand. Rather, it likely reflects unobserved pricing behavior or demand conditions across customer segments. Additional data on factors such as pricing decisions, remaining seat capacity, promotions and route characteristics would be needed to better understand what drives these differences.

Refer to Figure 10 in the Appendix for the relevant results generated using Python.

3.3 Lead Time Effects

Model 1 also shows that demand varies across booking stages. Relative to bookings made 0–2 days before departure, the estimated coefficients are +0.099 for 3–7 days, −0.049 for 8–14 days, −0.120 for 15–30 days, and −0.094 for >30 days. The results indicate that recorded demand differs across booking stages, supporting the inclusion of lead-time controls in Model 1.

Together with the change in the price coefficient from −0.207 in Model 0 to −0.253 in Model 1, this further supports the inclusion of categorical lead-time controls.
Refer to Figure 11 in the Appendix for the relevant results generated using Python.

3.4 Aggregation Sensitivity through Price Binning

Test 2 estimates Model 1 using $10, $20, $50 and $100 price bins while retaining the same lead time controls. The estimated price coefficients are −0.289, −0.307, −0.326 and −0.304, respectively, compared with −0.253 in Model 1, and all remain statistically significant ($p < 0.001$). The negative sign is robust across alternative price groupings, although the magnitude is somewhat sensitive to aggregation. Model 1 remains the preferred specification because it retains exact observed price variation and avoids imposing arbitrary price-bin aggregation.
Refer to Figure 12 in the Appendix for the relevant results generated using Python.

3.5 Diagnostic and Sensitivity Tests

The Breusch–Pagan test indicates unequal error variance ($p < 0.001$). Therefore, HC3 robust standard errors are used to improve the reliability of statistical inference.
Additional sensitivity checks were conducted to validate the divergence between customer categories:

- Test 1A estimates the segmented customer models without lead time controls yielded coefficients of +0.278 for Category A and −0.264 for Category B). Because Customer A remains positive and Customer B remains negative, the sign difference is not created by the five lead-time controls.
- Test 1B examines the influence of the unusual high leverage observation identified for Customer B by excluding the associated rare train observations (Train O). The price coefficient remains unchanged at −0.296, suggesting that this observation does not drive the Customer B point estimate.
  Refer to Figure 14 in the Appendix for the relevant results generated using Python.

3.6 Endogeneity and Limitations

While this model rigorously controls for observable fixed effects, the data is fundamentally observational. The estimates represent conditional associations rather than pure causal effects as modern rail ticket algorithms might execute dynamic pricing, raising prices in response to unobserved variables like remaining physical seat capacity or real time demand signals. Customer A's positive price coefficient is consistent with this. Additionally, simultaneity bias remains a limitation in perfectly isolating the demand curve. Notably, although HC3 robust standard errors improve inference when error variance is unequal, they do not address price endogeneity.

4. Conclusion

This project estimates the price elasticity of train travel demand while controlling for confounding effects of booking timing,trip details, customer category and departure timing. We demonstrated that booking timing is an important factor when estimating the price demand relationship. The estimated price coefficient is closer to zero when booking-stage controls are omitted, shifting from -0.207 in the baseline model to -0.253 after lead-time controls are included. Furthermore, splitting the sample reveals substantial heterogeneity across customer categories, with Customer B showing a negative price association (-0.296), compared with a positive association for Customer A (+0.165) . The negative pooled price association remains robust across alternative price-binning specifications, although its magnitude is somewhat sensitive to aggregation, while the Customer B estimate remains unchanged after excluding the rare high-leverage observation.. However, because ticket prices may dynamically respond to expected demand and other unobserved factors, these estimates should be interpreted as correlations rather than causal effects.

References:

1. Yuan, Y., Ran, W., & Zhang, S. (2026). Analysis of advance purchase behavior of air–rail passengers with ticket booking data. *Applied Sciences*, *16(13)*, Article 6300. https://doi.org/10.3390/app16136300
2. Hotle, S., Castillo, M., Garrow, L., & Higgins, M. (2015). The impact of advance purchase deadlines on airline consumers' search and purchase behaviors. *Transportation Research Part A: Policy and Practice*, *82*(1), 1 - 16. https://doi.org/10.1016/j.tra.2015.09.001

Appendix

Table 1. Choice of the lead-time window L

L (days)	Departure dates retained	Earliest departure	Records	Booking curve covered
60	335	2018-07-31	113,042	62.9%
90	305	2018-08-30	124,570	74.7%
120 (adopted)	275	2018-09-29	123,991	82.7%
180	215	2018-11-28	106,626	92.6%
240	155	2019-01-27	79,660	97.7%

Note. "Booking curve covered" is the share of a departure's bookings made within L days of departure, estimated from departures whose entire selling period falls inside the purchase window.

Figure 1

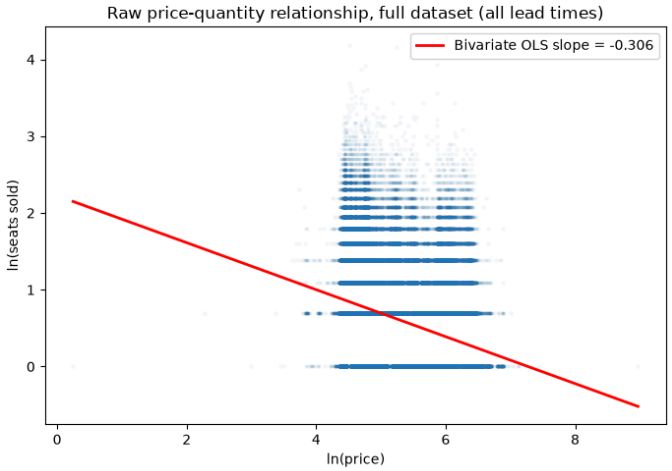


Figure 2

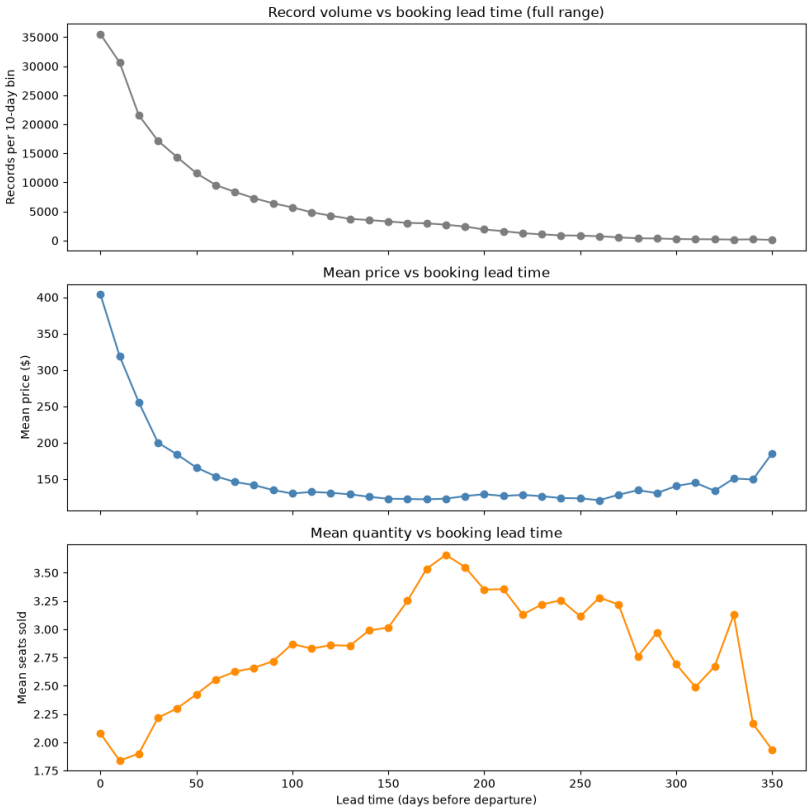


Figure 3

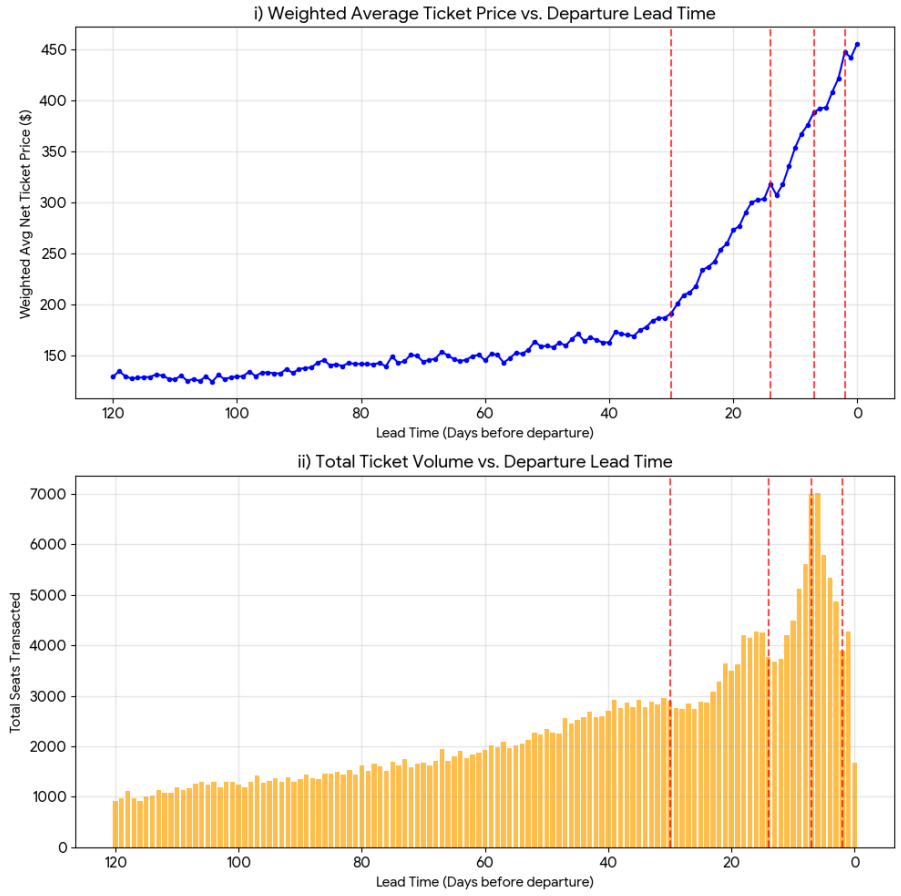


Figure 4

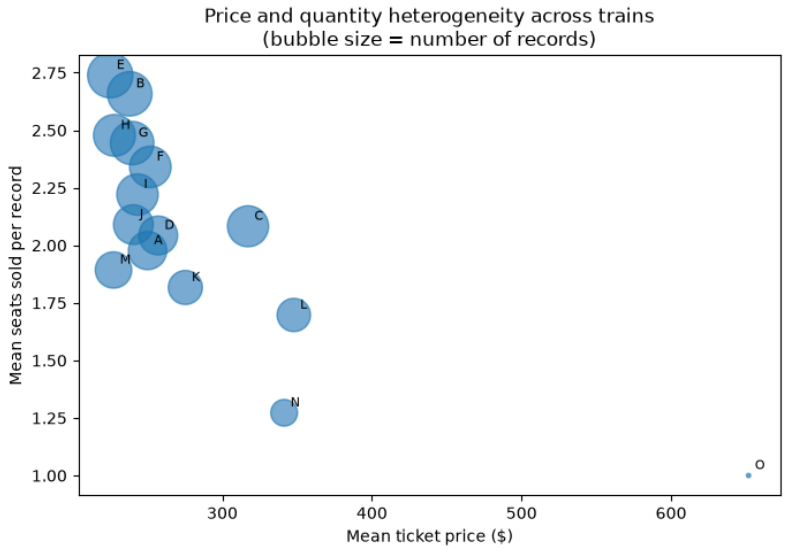


Figure 5

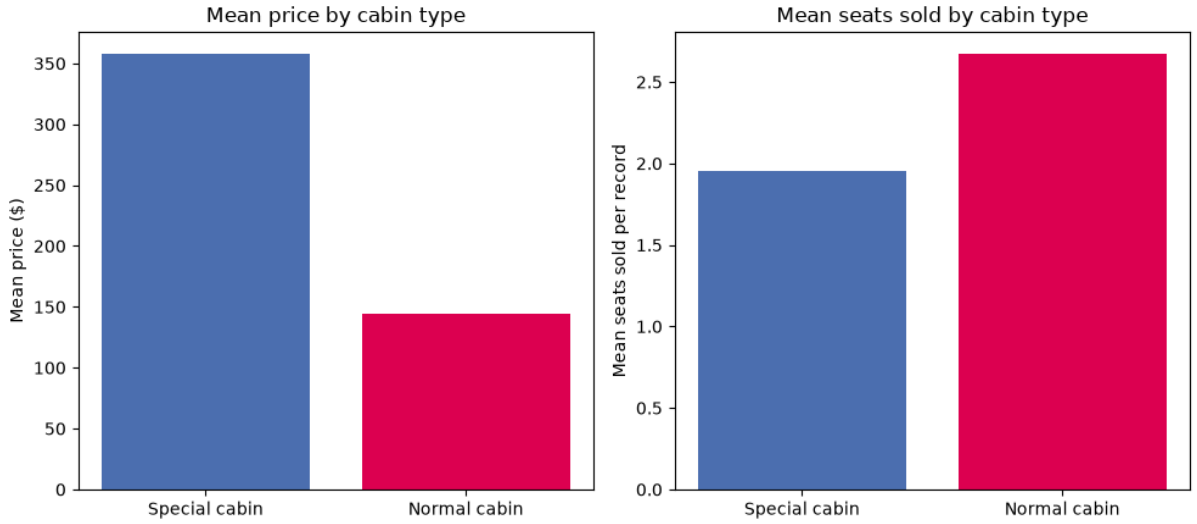


Figure 6

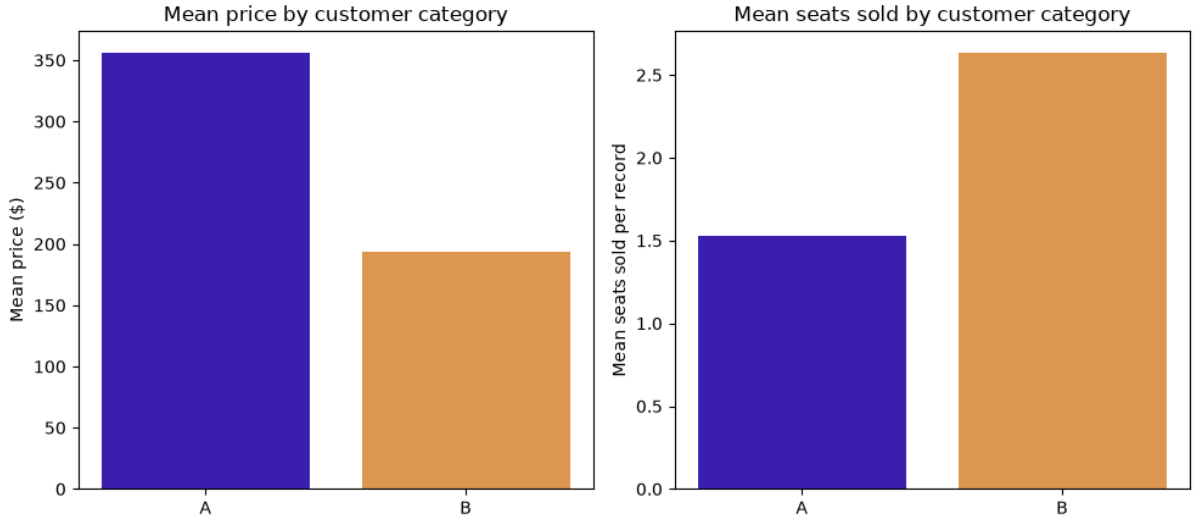


Figure 7

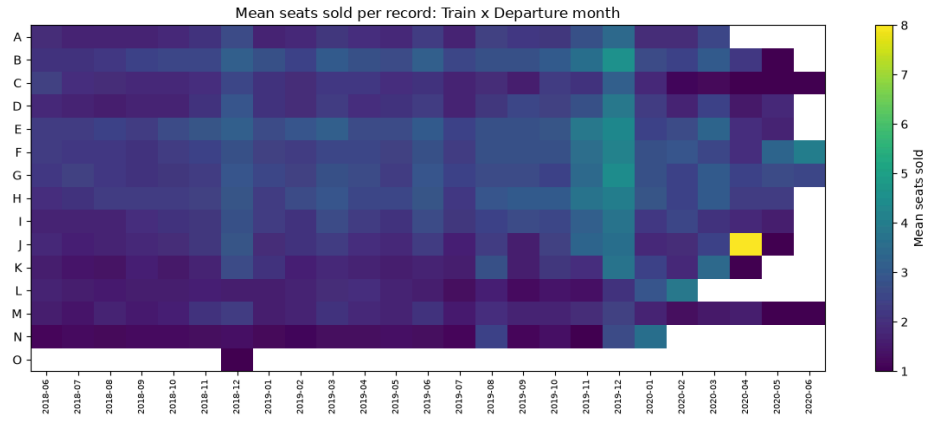


Figure 8

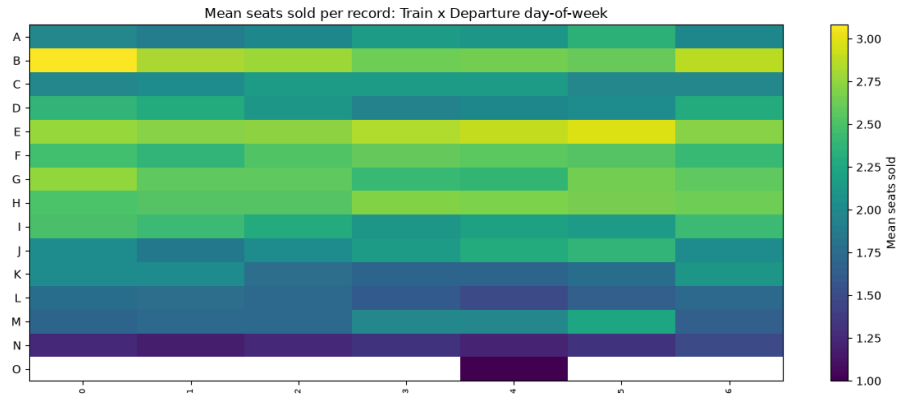


Figure 9

	Model	ln(price) coefficient	HC3 SE	95% CI lower	95% CI upper	p-value	N	R-squared	Adjusted R-squared
0	Reference Model 0: reference without lead-time...	-0.206706	0.004721	-0.215959	-0.197453	< 0.001	123991	0.128123	0.127884
1	Model 1: Main specification, exact observed price	-0.252765	0.005243	-0.263042	-0.242489	< 0.001	123991	0.136150	0.135885

Figure 10

	Customer category	Reference coefficient (without lead)	Main category coefficient (with lead)	Coefficient change (with minus without)	HC3 SE without lead	HC3 SE with lead	95% CI lower without	95% CI upper without	95% CI lower with	95% CI upper with	p-value without	p-value with	N without	N with	R-squared without	R-squared with	Adjusted R-squared without	Adjusted R-squared with
0	A	0.277995	0.164636	-0.113359	0.010711	0.010829	0.257003	0.298988	0.143412	0.185861	< 0.001	< 0.001	32401	32401	0.071044	0.095457	0.070126	0.094451
1	B	-0.264210	-0.295916	-0.031706	0.005181	0.005928	-0.274364	-0.254056	-0.307535	-0.284296	< 0.001	< 0.001	91590	91590	0.083747	0.088907	0.083417	0.088539

Figure 11

	Variable	Coefficient
0	3–7 days	0.0990
1	8–14 days	-0.0491
2	15–30 days	-0.1198
3	>30 days	-0.0936
4	$\ln(P_1)$	-0.2528

Figure 12

	Specification	Price-bin width	Price grouping	Source N	Aggregation reduction %	Difference vs baseline	Regression N	ln(price) coefficient	HC3 SE	t/z-stat	p-value	95% CI lower	95% CI upper	R-squared	Adjusted R-squared	Design columns	Design rank	Residual degrees of freedom	Price separately estimable	Maximum leverage	Cells with leverage >= 0.99	Finite price HC3 SE
0	Model 1 baseline: exact observed price	NaN	Exact observed price	123991	0.000000	0.000000	123991	-0.252765	0.005243	-48.208127	< 0.001	-0.263042	-0.242489	0.136150	0.135885	39	39	123952.0	True	1.0	1	True
1	Test 2: $10 bins	10.0	$10 bins	123991	29.263414	-0.036671	87707	-0.289436	0.007016	-41.252675	< 0.001	-0.303188	-0.275685	0.216364	0.216025	39	39	87668.0	True	1.0	1	True
2	Test 2: $20 bins	20.0	$20 bins	123991	33.587115	-0.054536	82346	-0.307302	0.007431	-41.353021	< 0.001	-0.321867	-0.292737	0.208676	0.208311	39	39	82307.0	True	1.0	1	True
3	Test 2: $50 bins	50.0	$50 bins	123991	40.686824	-0.073630	73543	-0.326396	0.007999	-40.802847	< 0.001	-0.342074	-0.310717	0.204563	0.204152	39	39	73504.0	True	1.0	1	True
4	Test 2: $100 bins	100.0	$100 bins	123991	45.789614	-0.051394	67216	-0.304159	0.008415	-36.145782	< 0.001	-0.320652	-0.287667	0.203625	0.203174	39	39	67177.0	True	1.0	1	True

Figure 13

	Regression N	ln(price) coefficient	HC3 SE	t/z-stat	p-value	95% CI lower	95% CI upper	R-squared	Adjusted R-squared
Customer A: reference without lead controls	32401	0.277995	0.010711	25.954595	< 0.001	0.257003	0.298988	0.071044	0.070126

	Regression N	ln(price) coefficient	HC3 SE	t/z-stat	p-value	95% CI lower	95% CI upper	R-squared	Adjusted R-squared
Customer B: reference without lead controls	91590	-0.26421	0.005181	-50.996734	< 0.001	-0.274364	-0.254056	0.083747	0.083417

Figure 14

	Specification	N	ln(price) coefficient	HC3 SE	t/z-stat	p-value	95% CI lower	95% CI upper	R-squared	Adjusted R-squared	Maximum leverage	Observations with leverage >= 0.99
0	Test 1: Customer B	91590	-0.295916	0.005928	-49.915221	< 0.001	-0.307535	-0.284296	0.088907	0.088539	1.000000	1
1	Test 1B: Customer B excluding rare train	91589	-0.295916	0.005928	-49.915221	< 0.001	-0.307535	-0.284296	0.088896	0.088538	0.003583	0